

PH 724: Data Analysis & Scientific Communication Incorporating Use of Generative AI, Fall 2026

Credits: 3

Meeting times: Mon 11:30—14:10 @Zilber 110

Instructor

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Office hours: Mon 16:00—17:00 @Zilber 376

Welcome Statement

Welcome to *Data Analysis & Scientific Communication Incorporating Use of Generative AI*! I'm excited to have you in this course and look forward to helping you develop skills in data analysis, scientific communication, and the responsible application of generative AI tools. Together, we will explore how data-driven approaches can support research and decision-making across disciplines. My goal is to create an inclusive and collaborative learning environment where students of all backgrounds can ask questions, share ideas, and grow with confidence.

Course Overview

Statistical methods play a critical role in generating evidence and informing decisions across a wide range of scientific, medical, engineering, social, and business applications. Beyond methodological expertise, successful statisticians and data scientists must be able to collaborate effectively with domain experts, understand research objectives, and translate complex questions into meaningful analytical strategies. The overall objective of this course is to introduce students to the practice of statistical consulting and interdisciplinary data analysis, including problem formulation, study design, statistical analysis planning, interpretation of results, communication of findings, and development of reproducible analytical workflows. This course combines lectures, literature discussions, case studies, and presentations to expose students to realistic consulting scenarios. Students will learn how to address common challenges encountered in collaborative research, including defining objectives, determining appropriate data collection strategies, evaluating sample size and study design considerations, selecting statistical approaches, and communicating results to both technical and non-technical audiences. Through these activities, students will develop practical skills in project management, teamwork, scientific communication, and professional reporting.

Course Objectives

- Introduce principles and best practices for applying statistical methods to address scientific and data-driven questions across diverse disciplines.

- Develop students' ability to approach real-world research problems through effective study design, data management, analytical reasoning, and interpretation of results.
- Enhance students' skills in communicating quantitative information and research findings to diverse audiences through written, oral, and visual formats.
- Familiarize students with modern computational tools, data visualization approaches, and generative AI technologies that support efficient, reproducible, and responsible data analysis workflows.
- Promote critical thinking in evaluating statistical methodologies, research evidence, and the appropriate use of analytical and AI tools in scientific investigations.

Student Learning Outcomes

- By the end of the course, students will be able to:
- Perform all responsibilities of a statistician in collaborative research; in particular: design studies, manage and analyze data and interpret findings from a variety of experimental and observational studies.
- Communicate statistical information effectively with individuals with varying degrees of statistical knowledge through written and oral presentations.
- Use computing software and AI tools to organize, analyze, and visualize data.
- Review and critique statistical methods and interpretation of results in published research studies, presentations, or reports.

Assessment

Students will be evaluated through regular assignments and final project. Assessment is aligned with the stated learning outcomes and emphasizes reproducibility, practical proficiency, and clear data communication.

Prerequisites and/or Special Skills Required

Graduate student; PH711/PH759 and PH718 or consent of instructor.

Course Modality and Format

- Format: Lectures.
- Modality: Onsite.
- Detail: Active participation during class time is essential, as learning will take place through live instruction, hands-on coding, group discussions, and collaborative problem-solving.
- Interaction with the instructor: The instructor will lead live lectures and demonstrations during each class session, introducing new concepts and walking through code trunks in real time. Students are expected to ask questions, share screens when needed, and engage in discussion during class. Office hours (by appointment) will be held on lecture days for one-on-one support or deeper technical questions.
- Interaction with peers: Students will work in breakout groups, pairs, or small teams during class to complete coding challenges, troubleshoot data issues, and review visualizations. Respectful, professional interaction is expected at all times, whether in discussion or screen-sharing activities.

- Interaction with course content: Lecture notes and code examples will be posted before each lecture to support self-directed learning. Students are expected to review these materials in advance so that live sessions can focus on hands-on coding, clarification, and collaborative exercises. Course content is reinforced through regular assignments and in-class application.

Course Materials

- Required reading:
 - Notes/slides regularly posted at [the instructor's homepage](#) and Canvas.
 - P. McCullagh. (2022). *Ten Projects in Applied Statistics*. New York: Springer. (Accessible author's copy at <https://www.stat.uchicago.edu/~pmcc/projects/>.)
 - [P#1] Kass, R. E., Caffo, B. S., Davidian, M., Meng, X. L., Yu, B., & Reid, N. (2016). Ten simple rules for effective statistical practice. *PLoS Computational Biology*, 12(6), e1004961.
 - [P#2] Fernandes-Taylor, S., Hyun, J. K., Reeder, R. N., & Harris, A. H. (2011). Common statistical and research design problems in manuscripts submitted to high-impact medical journals. *BMC Research Notes*, 4(1), 304.
 - [P#3] Barkan, H. (2015). Statistics in clinical research: Important considerations. *Annals of Cardiac Anaesthesia*, 18(1), 74.
 - [P#4] Habibzadeh, F. (2017). How to report the results of public health research. *Journal of Public Health and Emergency*, 1(12).
 - [P#5] Cao, Y., Chen, R. C., & Katz, A. J. (2024). Why is a small sample size not enough? *The Oncologist*, 29(9), 761-763.
 - [P#6] Button, K. S., Ioannidis, J. P., Mokrysz, C., Nosek, B. A., Flint, J., Robinson, E. S., & Munafò, M. R. (2013). Power failure: why small sample size undermines the reliability of neuroscience. *Nature Reviews Neuroscience*, 14(5), 365-376.
 - [P#7] Greenland, S., & Finkle, W. D. (1995). A critical look at methods for handling missing covariates in epidemiologic regression analyses. *American Journal of Epidemiology*, 142(12), 1255-1264.
 - [P#8] Steyerberg, E. W., Vickers, A. J., Cook, N. R., Gerds, T., Gonen, M., Obuchowski, N., ... & Kattan, M. W. (2010). Assessing the performance of prediction models: a framework for some traditional and novel measures. *Epidemiology (Cambridge, Mass.)*, 21(1), 128.
 - [P#9] Mayén, A. L., Guessous, I., Paccaud, F., Stringhini, S., & Marques-Vidal, P. (2016). Educational differences in dietary intake and compliance with dietary recommendations in a Swiss adult population. *International Journal of Public Health*, 61(9), 1059-1067.
- Recommended reading:
 - J. Derr (2000). *Statistical Consulting: A Guide to Effective Communication*. Duxbury Press.
 - J. Cabrera & A. McDougall (2001). *Statistical Consulting*. New York: Springer.
 - G. James, D. Witten, T. Hastie, & R. Tibshirani. (2021). *An Introduction to Statistical Learning: with Applications in R*, 2nd Ed. New York: Springer. (Accessible at www.statlearning.com.)

- R. Leung (2025). *Leveraging Gen AI for Machine Learning Education in Public Health*. Cham: Springer Nature. (Accessible e-copy via the [UWM library](#).)

Time Investment

This is 3-credit course, which requires a minimum of 144 hours of total time commitment over the semester. This translates to approximately 9–10 hours of work per week across 15 weeks, including both in-class and out-of-class activities. Each week, students should expect to spend their time roughly as follows: 2.5 hours in lectures, 2–3 hours reviewing pre-posted lecture materials, reading documentation, or exploring example code, and 4.5 hours completing assignments or working on final projects.

Technology Requirements

Each student is expected to have regular access to a computer with a reliable internet connection. This course makes extensive use of software packages for statistical computing.

Assignments & Grading

Assignment and Grading Policies

- Assignments will relate to readings, lectures, class discussions, and project reports. Some will be completed in class and others will be take-home assignments. The purpose of assignments is to provide students an opportunity to familiarize themselves with the principle of biostatistical consulting. The assignments will help students learn to understand the statistical issues related to a scientific investigation, apply appropriate statistical methods for data analysis, and effectively communicate with clients. Project reports are expected to be turned in on time. Five points will be deducted each day for late written report, unless there are extenuating circumstances. There may be rare exceptions made to this policy on an individual basis, provided that this is worked out in advance with the instructor.
- The final grade will be based on the weighted average of scores of the student-led discussion on selected statistical papers and the project oral presentation and written report. Both technical content and the presentation of that content are important. The first project will be down-weighted as it is considered as an initial learning exercise. Active participation in class discussions is important for success in the class.

Grading Scheme

Assessment (Quantity)	Point Value
Projects 1–4	60
Final individual project (1)	40
Total	100

Grading Scale

Grade	Scale
A	93–100
A-	90–92
B+	87–89
B	83–86
B-	80–82
C+	77–79
C	73–76
C-	70–72
D+	67–69
D	63–66
D-	60–62
F	0–59

Policies

Course Policies

- Attendance Policies: Though there is no penalty on absence, it is better to be present in the designated session punctually. Since the course will be delivered without any forms of recording, there is no alternative way of attendance.
- Makeup or Late work. No late submissions will be accepted. Timely submissions will be graded and returned within two weeks. Students with personal or medical emergencies should contact their instructor as soon as possible.
- Academic Integrity Policy. Academic integrity involves taking responsibility for one's own work, demonstrating honesty in academic efforts, and showing respect for the intellectual contributions of others. As members of the university community, all students are expected to understand and uphold these principles. The Joseph J. Zilber College of Public Health remains committed to maintaining high standards of academic integrity. Each student is expected to contribute to this commitment by avoiding academic misconduct, including but not limited to impersonation, plagiarism, and the use of unauthorized materials. In situations where expectations are unclear (such as whether collaboration, tools, or specific resources are permitted), seeking clarification from the instructor is encouraged. Asking questions reflects a commitment to integrity and a desire to engage responsibly with the academic process. Useful resources can be found at UWM [information on academic misconduct](#). Also, please refer to [CETL's tips on academic integrity in online learning for instructors and students](#).
- Policy on Artificial Intelligence. Students are permitted to use Generative Artificial Intelligence (GenAI) tools, such as [Microsoft Copilot](#), to support their learning. These tools may assist with understanding course content, brainstorming ideas, and revising written drafts. However, final submissions for assignments and projects must reflect students' own work and should not be entirely generated by GenAI. If GenAI tools are used, proper citation is required. For guidance on citing AI-generated content, please refer to the resource on [using and citing GenAI](#), provided by UWM Libraries. Taking credit for any work

that is not original, whether in whole or in part, constitutes academic misconduct and may lead to disciplinary action under [UWS Chapter 14](#). While GenAI can be helpful, it is essential to use them responsibly and ethically. AI-generated content may be biased, inaccurate, incomplete, or otherwise unreliable. Students must critically evaluate any output from GenAI before incorporating it into academic work.

University Policies

- Statement on Hate/Bias Incidents: UWM is committed to creating and supporting a campus climate that is respectful and supportive of all who study, live, or work on campus, or participate in campus activities. UWM students and employees who have been subjected to, or who have witnessed, an incident of bias or hate are encouraged to submit a [hate/bias incident report](#). UWM can provide support or resources to those involved in the incident. Find more information about hate- or bias-motivated incidents, as well as UWM's Discriminatory Conduct Policy and other resources, on UWM's [hate/bias webpage](#).
- UWM Syllabus link: The [Syllabus Links](#) page details policies pertaining to students with disabilities, absences due to religious observation, students called to active military duty, incompletes, discriminatory conduct, Title IX, academic misconduct, complaint procedures, grade appeal procedures, LGBT+ resources, and final exam policies.

Resources

Academic Supports

- [Accessibility Resource Center](#). Dedicated to providing equal access for students with disabilities to UWM's academic, social, cultural and recreational programs.
- Canvas Support: Visit [UWM Canvas Student Support](#) or reference the official [Canvas Student Guide](#).
- [Student Success Center](#): Explore tutoring options for your courses or attend a supplemental instruction session. Student Success Center also offers one-on-one academic skills tutoring sessions to help students with time management and strategies for notetaking, studying, and test taking.
- [Textbooks via Panther Shop](#). Find out what materials you need for your courses.
- [UWM Libraries](#). Conduct your own research using a vast array of databases, seek help from librarians, reserve rooms for group projects, find a quiet study space, stop by the coffee shop.
- [Writing Center](#). Meet confidentially, one-on-one, face-to-face or online in real time with writing specialists. Any subject, any project, any level.

Non-Academic Supports

- [Support U](#): Any student in need, or students that face challenges that are barriers to their education, are encouraged to contact the Dean of Students (dos@uwm.edu) for support. Support U offers wrap-around holistic support for students, including basic needs, accessing the food pantry, emergency funding, case management, and connecting to resources, etc.
- [Emergencies](#): Includes crisis and after care, details on medical withdrawals, and UWM Police.

- [Financial Resources](#): Links to Student Financial Services, scholarship information, emergency grants, and the food pantry.
- [Health Resources](#): Medical and counseling services, community resource finder, the UWM Psychology clinic, survivor support and victim advocacy.
- [Guidance and Consultation](#): Details on appeals, parking, accommodations, and mechanisms for filing grievances.

Technology Resources

- [Student Technology Resources](#)

Careers

- [Student Experience and Talent](#)

Centers for Advocacy & Student Engagement (CASE)

- [First-Generation+ Resource Center](#)
- [LGBTQ+ Resource Center](#)
- [Military and Veterans Resource Center](#)
- [Women's Resource Center](#)

Multicultural Student Centers (MSCs)

- [American Indian Student Center](#)
- [Black Student Cultural Center](#)
- [Roberto Hernández Center](#)
- [Southeast Asian American Student Center](#)

Student Involvement

- [Event Calendar](#)
- [Student Association \(Student Government\)](#)
- [Student Involvement/Organizations](#)
- [Office of Undergraduate Research](#)

Important Dates

Event

First day of classes
 Labor Day
 Add deadline
 Last day to drop without a "W"
 Drop deadline
 No class (Thanksgiving)
 Last day of classes
 Study Day
 Final exams begin
 Grade Deadline

Date

Sept. 2, 2026
 Sept. 7, 2026
 Sept. 9, 2026
 Sept. 29, 2026
 Nov. 8, 2026
 Nov. 25–29, 2026
 Dec. 14, 2026
 Dec. 15, 2026
 Dec. 16, 2026
 Jan. 4, 2027

Instructor's statement of support

I know this course may look challenging at first glance. Learning to work with data takes time, practice, and persistence. Please don't hesitate to reach out if you're struggling, stuck, or just need a nudge in the right direction. I believe each of you can succeed in this class, and I'm committed to helping you get there. Let's work together to make this a productive and supportive semester.

Calendar

Date	Topic	Papers
Sept. 14	Syllabus review and beyond	
Sept. 21	Introduction to Project 1 (Rat Surgery)	[P#1] [P#2]
Sept. 28	Presenting preliminary results of Project 1	
Oct. 5	Presenting final reports of Project 1	
Oct. 12	Introduction to Project 2 (Growth Curves)	[P#3] [P#4]
Oct. 19	Presenting preliminary results of Project 2	
Oct. 26	Presenting final reports of Project 2	
Nov. 2	Introduction to Project 3 (Environmental Projects)	[P#5] [P#6]
Nov. 9	Presenting preliminary results of Project 3	
Nov. 16	Presenting final reports of Project 3	
Nov. 23	Introduction to Project 4 (Out of Africa)	[P#7] [P#8] [P#9]
Nov. 30	Presenting preliminary results of Project 4	
Dec. 7	Presenting final reports of Project 4	
Dec. 14	Presentation for final projects	